

99.500 Applied Mathematics for Engineering

Homework 4 Solution

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1. The general rate equation for any protein species i is

$$\frac{dC_i}{dt} = k_{i-1}C_{i-1} - k_iC_i \text{ with } i = 1, 2, \dots, N \quad (1)$$

and C_i denotes the concentration of P_i . Here, when $i = 1$, $k_0 = 0$ and when $i = N$, $k_N = 0$. Take the Laplace transform of above equation,

$$\mathcal{L}\left(\frac{dC_i}{dt}\right) = \mathcal{L}(k_{i-1}C_{i-1} - k_iC_i) \quad (2)$$

to obtain

$$(s + k_i)\overline{C}_i(s) - C_i(0) = k_{i-1}\overline{C}_{i-1}(s) \quad (3)$$

$$i = 1 \text{ \& } C_1(0) = C_{\text{initial}} \implies \overline{C}_1(s) = \frac{C_{\text{initial}}}{s + k_1} \quad (4)$$

$$i = 2, 3, \dots, N-1 \text{ \& } C_i(0) = 0 \implies \overline{C}_i(s) = \frac{k_{i-1}\overline{C}_{i-1}(s)}{s + k_i} \quad (5)$$

$$i = N \implies \overline{C}_N(s) = \frac{k_{N-1}\overline{C}_{N-1}(s)}{s} \quad (6)$$

From Equation (5) to have

$$\overline{C}_i(s) = \frac{k_{i-1}\overline{C}_{i-1}(s)}{s + k_i} = \frac{k_{i-1}}{s + k_i} \frac{k_{i-2}}{s + k_{i-1}} \overline{C}_{i-2}(s) = \frac{C_{\text{initial}}}{s + k_1} \prod_{j=2}^i \frac{k_{j-1}}{s + k_j} \quad (7)$$

Express $\overline{C}_i(s)$ in the partial fraction of the form

$$\overline{C}_i(s) = \left(C_{\text{initial}} \prod_{j=2}^i k_{j-1} \right) \sum_{m=1}^i \frac{x_m}{s + k_m} \quad (8)$$

where x_m is the coefficient to be determined.

$$\frac{C_{\text{initial}}}{s + k_1} \prod_{j=2}^i \frac{k_{j-1}}{s + k_j} = \left(C_{\text{initial}} \prod_{j=2}^i k_{j-1} \right) \sum_{m=1}^i \frac{x_m}{s + k_m} \quad (9)$$

$$\implies \prod_{j=1}^i \frac{1}{s + k_j} = \sum_{m=1}^i \frac{x_m}{s + k_m} = \frac{\sum_{m=1}^i x_m \prod_{n=1, n \neq m}^i (s + k_n)}{\prod_{m=1}^i (s + k_m)} \quad (10)$$

$$\implies \sum_{m=1}^i x_m \prod_{n=1, n \neq m}^i (s + k_n) = 1 \quad (11)$$

When $s = -k_m$ in Equation (11),

$$x_m \prod_{n=1, n \neq m}^i (k_n - k_m) = 1 \quad (12)$$

Hence,

$$x_m = \frac{1}{\prod_{n=1, n \neq m}^i (k_n - k_m)} \quad (13)$$

$$\overline{C}_i(s) = C_{\text{initial}} \prod_{j=2}^i k_{j-1} \left[\sum_{m=1}^i \frac{1}{s + k_m} \frac{1}{\prod_{n=1, n \neq m}^i (k_n - k_m)} \right] \quad (14)$$

$$C_i(t) = \mathcal{L}^{-1}(\overline{C}_i(s)) = C_{\text{initial}} \prod_{j=2}^i k_{j-1} \left[\sum_{m=1}^i e^{-k_m t} \prod_{n=1, n \neq m}^i \frac{1}{(k_n - k_m)} \right] \quad (15)$$

The concentrations for P_1 , P_2 and P_3 can be evaluated from Equation (15) as

$$C_1(t) = C_{\text{initial}} e^{-k_1 t} \quad (16)$$

$$C_2(t) = \frac{C_{\text{initial}} k_1}{k_1 - k_2} (e^{-k_2 t} - e^{-k_1 t}) \quad (17)$$

$$C_3(t) = C_{\text{initial}} k_1 k_2 \left(\frac{e^{-k_1 t}}{(k_2 - k_1)(k_3 - k_1)} + \frac{e^{-k_2 t}}{(k_1 - k_2)(k_3 - k_2)} + \frac{e^{-k_3 t}}{(k_1 - k_3)(k_2 - k_3)} \right) \quad (18)$$

2. Assume $T(r, z) = \theta(r)\eta(z)$ and substitute into

$$\frac{\partial^2 T}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = 0 \quad (19)$$

to obtain

$$\frac{\eta''}{\eta} = -\frac{\theta'' + \frac{\theta'}{r}}{\theta} = -\lambda^2 \quad (20)$$

For $\theta(r)$ part:

$$r^2 \theta'' + r \theta' - \lambda^2 r^2 \theta = 0 \implies \theta = A_1 I_0(\lambda r) + A_2 K_0(\lambda r) \quad (21)$$

$$\left. \frac{\partial T}{\partial r} \right|_{r=0} = 0 \implies \theta'(0) = 0 \implies A_2 = 0 \implies \theta = A_1 I_0(\lambda r) \quad (22)$$

For $\eta(z)$ part:

$$\eta'' + \lambda^2 \eta = 0 \implies \eta = B_1 \sin(\lambda z) + B_2 \cos(\lambda z) \quad (23)$$

$$T(r, 0) = 0 \implies \eta(0) = 0 \implies B_2 = 0 \quad (24)$$

$$\left. \frac{\partial T}{\partial z} \right|_{z=\alpha} = 0 \implies \eta'(\alpha) = 0 \implies B_1 \lambda \cos(\lambda \alpha) = 0 \quad (25)$$

$$\implies \lambda_n = \left(n - \frac{1}{2} \right) \frac{\pi}{\alpha}, \quad n = 1, 2, 3, \dots \quad (26)$$

Hence,

$$T(r, z) = \sum_{n=1}^{\infty} \theta_n(r) \eta_n(z) = \sum_{n=1}^{\infty} C_n I_0 \left[\left(n - \frac{1}{2} \right) \frac{\pi r}{\alpha} \right] \sin \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] \quad (27)$$

$$\left. \frac{\partial T}{\partial r} \right|_{r=1} = f(z) \implies \sum_{n=1}^{\infty} C_n \left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} I_1 \left[\left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} \right] \sin \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] = f(z) \quad (28)$$

By using the formula

$$A_n = \frac{\int_a^b f(x) \phi_n(x) w(x) dx}{\int_a^b \phi_n^2(x) w(x) dx} \quad (29)$$

where $A_n = C_n \left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} I_1 \left[\left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} \right]$ to obtain

$$C_n \left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} I_1 \left[\left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} \right] = \frac{\int_0^\alpha f(z) \sin \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] dz}{\int_0^\alpha \sin^2 \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] dz} \quad (30)$$

Since

$$\int_0^\alpha \sin^2 \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] dz = \frac{\alpha}{2} \quad (31)$$

$$C_n = \frac{\int_0^\alpha f(z) \sin \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] dz}{\left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} I_1 \left[\left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} \right] \int_0^\alpha \sin^2 \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] dz} = \frac{2 \int_0^\alpha f(z) \sin \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] dz}{\pi \left(n - \frac{1}{2} \right) I_1 \left[\left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} \right]} \quad (32)$$

Hence, the final solution for $T(r, z)$ is

$$T(r, z) = \sum_{n=1}^{\infty} \frac{2 \int_0^\alpha f(z) \sin \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] dz}{\pi \left(n - \frac{1}{2} \right) I_1 \left[\left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} \right]} I_0 \left[\left(n - \frac{1}{2} \right) \frac{\pi r}{\alpha} \right] \sin \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] \quad (33)$$

When $f(z) = 1$,

$$\int_0^\alpha \sin \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] dz = \frac{\alpha}{\pi \left(n - \frac{1}{2} \right)} \quad (34)$$

$$T(r, z) = \sum_{n=1}^{\infty} \frac{2\alpha}{\pi^2 \left(n - \frac{1}{2} \right)^2 I_1 \left[\left(n - \frac{1}{2} \right) \frac{\pi}{\alpha} \right]} I_0 \left[\left(n - \frac{1}{2} \right) \frac{\pi r}{\alpha} \right] \sin \left[\left(n - \frac{1}{2} \right) \frac{\pi z}{\alpha} \right] \quad (35)$$

3. Non-dimensionalization: $\theta = \frac{T-T_0}{T_0}$, $\rho = \frac{r}{r_2}$ and $\eta = \frac{z}{L}$ to transform the governing equation with IC and BCs into

$$\frac{1}{\gamma^2} \frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \theta}{\partial \rho} \right) = 0 \quad \gamma = \frac{L}{r_2} \quad (36)$$

$$\left. \frac{\partial \theta}{\partial \rho} \right|_{\rho=0} = 0 \quad (37)$$

$$\left. \frac{\partial \theta}{\partial \rho} \right|_{\rho=1} = 0 \quad (38)$$

$$\theta(\rho, 0) = 0 \quad (39)$$

$$\left. \frac{\partial \theta}{\partial \eta} \right|_{\eta=1} = \begin{cases} \frac{\beta L}{T_0} & \rho \leq \zeta \\ 0 & \zeta < \rho < 1 \end{cases} \quad \zeta = \frac{r_1}{r_2} \quad (40)$$

Assume $\theta(\eta, \tau) = R(\rho)Z(\eta)$ and substitute into Equation (36) to have

$$\frac{Z''}{\gamma^2 Z} = -\frac{1}{\rho R} \frac{d(\rho R')}{d\rho} = \lambda^2 \quad (41)$$

For $R(\rho)$ part:

$$\rho^2 R'' + \rho R' + \lambda^2 \rho^2 R = 0 \implies R = A_1 J_0(\lambda \rho) + A_2 Y_0(\lambda \rho) \quad (42)$$

$$\left. \frac{\partial \theta}{\partial \rho} \right|_{\rho=0} = 0 \implies R'(0) = 0 \implies A_2 = 0 \implies R = A_1 J_0(\lambda \rho) \quad (43)$$

$$\left. \frac{\partial \theta}{\partial \rho} \right|_{\rho=1} = 0 \implies R'(1) = 0 \implies J_1(\lambda) = 0 \implies J_1(\lambda_n) = 0 \quad (44)$$

$$\text{where } \lambda_n \text{ is the } n\text{-th root of } J_1(x) = 0 \quad (45)$$

When $\lambda_0 = 0$, $J_0(\lambda_0 \rho) = 1$ which is non-trivial. Hence, the case $n = 0$ has to be included.

For $Z(\eta)$ part:

$$Z'' = \lambda^2 \gamma^2 Z \implies Z = B_1 \sinh(\lambda \gamma \eta) + B_2 \cosh(\lambda \gamma \eta) \quad (46)$$

$$\theta(\rho, 0) = 0 \implies Z(0) = 0 \implies B_2 = 0 \implies Z = B_1 \sinh(\lambda \gamma \eta) \quad (47)$$

A combination of $R(\rho)$ and $Z(\eta)$ gives

$$\theta(\rho, \eta) = \sum_{n=0}^{\infty} R(\rho)Z(\eta) = \sum_{n=0}^{\infty} C_n \sinh(\lambda_n \gamma \eta) J_0(\lambda_n \rho) \quad (48)$$

From Equation (40) to have

$$\left. \frac{\partial \theta}{\partial \eta} \right|_{\eta=1} = \sum_{n=0}^{\infty} C_n \lambda_n \gamma \cosh(\lambda_n \gamma) J_0(\lambda_n \rho) = \begin{cases} \frac{\beta L}{T_0} & \rho \leq \zeta \\ 0 & \zeta < \rho < 1 \end{cases} = f(\rho) \quad (49)$$

By using the formula

$$A_n = \frac{\int_a^b f(x) \phi_n(x) w(x) dx}{\int_a^b \phi_n^2(x) w(x) dx} \quad (50)$$

where $A_n = C_n \lambda_n \gamma \cosh(\lambda_n \gamma)$ to obtain

$$C_n \lambda_n \gamma \cosh(\lambda_n \gamma) = \frac{\int_0^1 f(\rho) J_0(\lambda_n \rho) \rho d\rho}{\int_0^1 J_0^2(\lambda_n \rho) d\rho} = \frac{\beta L}{T_0} \frac{\int_0^\zeta J_0(\lambda_n \rho) \rho d\rho}{\int_0^1 J_0^2(\lambda_n \rho) \rho d\rho} = \frac{2\zeta \beta L}{\lambda_n T_0} \frac{J_1(\lambda_n \zeta)}{J_0^2(\lambda_n)} \quad (51)$$

and

$$\theta(\eta, \tau) = \sum_{n=0}^{\infty} \frac{2\zeta \beta L}{\gamma \lambda_n^2 \cosh(\lambda_n \gamma) T_0} \frac{J_1(\lambda_n \zeta)}{J_0^2(\lambda_n)} \sinh(\lambda_n \gamma \eta) J_0(\lambda_n \rho) \quad (52)$$

However, the Equation (52) is not defined for the case $n = 0$ since $\lambda_0 = 0$. By evaluating the

limit for the Bessel and hyperbolic functions as

$$\begin{aligned}
& \lim_{\lambda_0 \rightarrow 0} \frac{2\zeta\beta L}{\gamma\lambda_0^2 \cosh(\lambda_0\gamma) T_0} \frac{J_1(\lambda_0\zeta)}{J_0^2(\lambda_0)} \sinh(\lambda_0\gamma\eta) J_0(\lambda_0\rho) \\
&= \frac{2\zeta\beta L}{\gamma T_0} \lim_{\lambda_0 \rightarrow 0} \frac{J_1(\lambda_0\zeta) \sinh(\lambda_0\gamma\eta) J_0(\lambda_0\rho)}{\lambda_0^2 \cosh(\lambda_0\gamma) J_0^2(\lambda_0)} \\
&= \frac{2\zeta\beta L}{\gamma T_0} \lim_{\lambda_0 \rightarrow 0} \frac{\frac{\lambda_0\zeta}{2} \cdot \lambda_0\gamma\eta}{\lambda_0^2 \cdot 1 \cdot 1} \\
&= \frac{2\zeta\beta L}{\gamma T_0} \cdot \frac{\zeta\gamma\eta}{2} \\
&= \frac{\zeta^2\beta L\eta}{T_0}
\end{aligned} \tag{53}$$

The the Equation (52) becomes

$$\theta(\eta, \tau) = \frac{\zeta^2\beta L\eta}{T_0} + \frac{2\zeta\beta L}{\gamma T_0} \sum_{n=1}^{\infty} \frac{J_1(\lambda_n\zeta)}{\lambda_n^2 \cosh(\lambda_n\gamma) J_0^2(\lambda_n)} \sinh(\lambda_n\gamma\eta) J_0(\lambda_n\rho) \tag{54}$$

In terms of $T(r, z)$,

$$\frac{T - T_0}{T_0} = \frac{r_1^2\beta z}{r_2^2 T_0} + \frac{2r_1\beta}{T_0} \sum_{n=1}^{\infty} \frac{J_1\left(\lambda_n \frac{r_1}{r_2}\right)}{\lambda_n^2 \cosh\left(\lambda_n \frac{L}{r_2}\right) J_0^2(\lambda_n)} \sinh\left(\lambda_n \frac{z}{r_2}\right) J_0\left(\lambda_n \frac{r}{r_2}\right) \tag{55}$$